Prayush B Menon

628-263-2031 | bmenonprayush@gmail.com | LinkedIn | GitHub | Portfolio

OBJECTIVE

Embedded Systems & Firmware Engineer with an ECE background and hands-on expertise in bare-metal C development on PIC microcontrollers. Proficient in serial communication protocols (CAN, UART, SPI, I2C), Linux systems programming (POSIX, IPC, signals), memory-efficient data structures, and GDB debugging. Experienced in low-level firmware architecture, hardware-software integration, and software test automation.

TECHNICAL SKILLS

Languages: C (C99), Embedded C, C++, Java
Microcontrollers & Protocols: PIC18F4580, CAN, UART, I2C, SPI, ADC
Systems & Concepts: POSIX System Calls, IPC (Pipes, Signals), Multiprocessing, Memory Management, Bit Manipulation
Developer Tools & OS: Linux (Fedora/Ubuntu), GDB, Make, Git, Neovim, Vim, Postman, Burp Suite
Testing & Infrastructure: Selenium WebDriver, Performance Monitoring (Lighthouse), Test Automation, JIRA

PROJECTS

Minishell (Unix Command Shell) GitHub

C, Linux Systems Programming, POSIX APIs, Signals, Process Management

- Built a Unix command shell in C supporting command execution, dynamic prompts, and multi-stage pipelines using `pipe()` and `dup2()`.
- Implemented signal handling for `SIGINT`, `SIGTSTP`, and `SIGCHLD`, with a linked-list job table to manage foreground and background processes.

Multithreaded TFTP Server & Client GitHub

C, UDP Sockets, POSIX Threads, Semaphores, Network Protocols

- Built a multithreaded TFTP file transfer server and client in C over UDP sockets, handling concurrent client uploads and downloads with POSIX threads.
- Limited active client transfers to 10 using a counting semaphore (`sem_trywait`) that sends instant busy replies, assigning each transfer to a dedicated worker thread and port.

Car Black Box (Event Data Recorder) GitHub

Embedded C, PIC18F4580, I2C, UART, ADC, EEPROM, RTC, LCD

- Wrote bare-metal firmware for a PIC18F4580 microcontroller to record vehicle speed, gear shifts, and clock timestamps to an external EEPROM over I2C.
- Implemented a circular buffer retaining the 10 most recent events across power resets, with drivers for an ADC speed potentiometer, matrix keypad, LCD menu, and UART serial export.

Car Dashboard (Automotive CAN Simulator) GitHub

Embedded C, PIC18F4580, CAN Protocol, LCD, Embedded Systems

- Built a two-node automotive dashboard simulator using PIC18F4580 microcontrollers communicating over a CAN bus.
- Transmitted real-time vehicle telemetry (speed, RPM, gear) from the engine node and decoded incoming CAN frames to update a live LCD dashboard.

Lexical Analyzer for C GitHub

C, Lexical Analysis, Tokenization, Binary Search, File I/O

- Built a C source code tokenizer that strips comments and preprocessor directives, classifying code into keywords, identifiers, constants, and operators.
- Used binary search over sorted keyword arrays for fast lookups, and added syntax checks for unclosed quotes and unmatched parentheses or braces.

Arbitrary Precision Calculator (APC) GitHub

C, Data Structures, Algorithms, Dynamic Memory Allocation

- Built an arbitrary-precision calculator in C using doubly linked lists to perform arithmetic on numbers of any size without 64-bit integer limits.
- Implemented addition, subtraction, multiplication, and division with manual carry and borrow logic across nodes, ensuring zero memory leaks.

Image Steganography (LSB) GitHub

C, Bit Manipulation, Binary File I/O, Image Processing

- Built a command-line tool in C to hide and extract text messages inside 24-bit BMP images using the Least Significant Bit (LSB) technique.
- Replaced the lowest bits of RGB pixel bytes with secret payload data while preserving the 54-byte BMP header to keep images visually unchanged.

MP3 Tag Reader & Editor GitHub

C, Binary File Parsing, ID3v2 Tags, File I/O

- Built a command-line tool in C to view and edit ID3v2.3 metadata (title, artist, album) directly inside MP3 files.
- Parsed binary frame headers, handled big-endian byte offsets, and updated tag text in-place without corrupting the audio bitstream.

Source2HTML (C Source to HTML Converter) GitHub

C, State Machines, Lexical Parsing, File I/O, HTML/CSS

- Built a C source code highlighter that scans files using a finite state machine and converts them into syntax-highlighted HTML.
- Mapped keywords, strings, and comments into CSS classes, escaping special HTML characters (`<`, `>`, `&`) for clean browser rendering.

EXPERIENCE

Senior Quality Engineer I July 2024 – January 2026

Fresh Prints Remote

- Built an automated frontend performance testing infrastructure with Lighthouse, reducing manual testing time by 80% through bi-weekly automated performance monitoring across environments.
- Led backend data migration validation for 7,000+ design records from Shopify to Directus CMS, developing custom validation scripts that ensured 100% data integrity and schema consistency.
- Identified critical P1 security vulnerabilities using Burp Suite to detect unauthorized access vectors, collaborating directly with development teams for immediate patch verification.

QA Automation Engineer June 2023 – July 2024

Cognizant Technology Solutions Bangalore, India

- Developed and maintained test automation frameworks using Java and Selenium WebDriver, standardizing regression testing across core application modules.
- Triaged and tracked software defects across agile sprint cycles, performing root-cause analysis to verify fixes and ensure release reliability.
- Reviewed technical requirements and system specifications to author automated test plans, expanding test coverage across end-to-end user workflows.
- Collaborated with engineering teams to optimize test execution pipelines, migrating manual validation suites into continuous automated test runs.

QA Engineer June 2022 – June 2023

Cognizant Technology Solutions Bangalore, India

- Executed functional and regression test suites across multi-tier applications, isolating defect root causes and reducing turnaround time for issue resolution.
- Performed REST API validation and endpoint testing using Postman, verifying payload schemas, HTTP status codes, and backend data flows.
- Authored and maintained comprehensive test suites covering product functional specifications, mitigating critical defect escapes into production releases.
- Scripted test automation routines to replace repetitive manual verification, accelerating regression testing cycles while maintaining consistent test repeatability.

EDUCATION & TRAINING

Emertxe Information Technologies Bangalore, India

Advanced Embedded Systems Training (PIC18F4580, C/C++, Protocols, Linux) 2025 – 2026

Amrita School of Engineering Kollam, Kerala

B.Tech in Electronics and Communications Engineering (CGPA: 8.1/10) June 2018 – August 2022